



FOOD SCIENCE AND TECHNOLOGY DRAFT SAMPLE EXAMINATION STAGE 2

Section 7 of the *WACE Manual: General Information 2008 Revised Edition* outlines the policy on WACE examinations.

Further information about the WACE Examinations policy can be accessed from the Curriculum Council website at <http://www.curriculum.wa.edu.au/>

The purpose for providing a sample examination is to provide teachers with an example of how the course will be examined. Further finetuning will be made to this sample in 2008 by the examination working party following consultation with teachers, measurement specialists and advice from the Assessment, Review and Moderation (ARM) panel.

DRAFT



Western Australian Certificate of Education Draft Sample Examination Question/Answer Booklet

FOOD SCIENCE AND TECHNOLOGY STAGE 2

Please place your student identification label in this box

Student Number: In figures

--	--	--	--	--	--	--	--

In words

Time allowed for this paper

Reading time before commencing work: Ten minutes
Working time for paper: Three hours

Material required/recommended for this paper

To be provided by the supervisor

This question/answer booklet
Multiple-choice answer sheet

To be provided by the candidate

Standard items: Pens, pencils, eraser, correction fluid, ruler, highlighter

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Suggested working time	Number of questions available	Number of questions to be attempted	% of paper	Marks
One: Multiple-choice	20 minutes	20	20	20%	20
Two: Short response	100 minutes	8	8	50%	50
Three: Extended answer	30 minutes	1	1	15%	15
Four: Context-specific extended answer	30 minutes	6	1	15%	15
Total marks					100

Instructions to candidates

- The rules for the conduct of Curriculum Council examinations are detailed in the *Student Information Handbook*. Sitting this examination implies that you agree to abide by these rules.
- Answer the questions according to the following instructions.
Section ONE Answer **ALL** questions in the spaces provided in the multiple-choice answer sheet.
Section TWO Answer **ALL** questions in the spaces provided in this question/answer booklet.
Section THREE Answer **ONE** question in the pages provided in this question/answer booklet.
Section FOUR Answer **ONE** question from the context you have studied, in the pages provided in this question/answer booklet.
- A blue or black ballpoint or ink pen should be used.

SECTION ONE—MULTIPLE-CHOICE

Attempt **ALL** 20 questions from this section. Choose the alternative, which correctly answers the question or completes the statement. Record your answers on the separate multiple-choice answer sheet. Read the instructions on this sheet with care.

Suggested working time: 20 minutes [20 marks].

1. Which variety of apples is most appropriate for use as a filling in a baked apple pie? [1 mark]
(A) Royal Gala
(B) Granny Smith
(C) Fuji
(D) Golden Delicious.
2. When cooking cereal products: [1 mark]
(A) cellulose softens and absorbs vitamins
(B) B group vitamins are retained
(C) starch grains burst and gelatinise
(D) liquid absorbs the calcium and iron.
3. The process of adding nutrients to processed foods such as adding vitamins to cereal is called: [1 mark]
(A) irradiation
(B) fermentation
(C) nutrification
(D) fortification.
4. When developing a new food product the first stage in the process is to: [1 mark]
(A) generate ideas
(B) develop a prototype
(C) investigate needs
(D) conduct a feasibility study.
5. Cross-contamination refers to: [1 mark]
(A) the production of toxins by bacteria
(B) the transfer of micro-organisms from one food to another
(C) the improper use of chemical pesticides
(D) the presence of insects in fruit and vegetables.
6. Freezing is a popular method of food preservation for the consumer today because: [1 mark]
(A) freezing maintains the nutritional value of all food products
(B) frozen food products can be defrosted and refrozen without damaging quality
(C) during power failure, unopened freezers will help prevent food deterioration
(D) product sensory qualities are quite well preserved.

7. Which of the following is **not** an example of a functional food?

[1 mark]

- (A) a beef burger made with added soy protein
- (B) a vegetarian lasagna made with spinach and zucchini
- (C) margarine made with phytosterols
- (D) a carbonated drink containing ginseng and Echinacea.

8. Which of the following food combinations is an excellent source of folate?

[1 mark]

- (A) cheese and green, leafy vegetables
- (B) green, leafy vegetables and legumes
- (C) legumes and milk
- (D) milk and meat.

9. Which of the following is a combination of both dry and moist heat cooking methods?

[1 mark]

- (A) braising
- (B) roasting
- (C) blanching
- (D) en papillote.

10. The Maillard reaction is:

[1 mark]

- (A) the browning reaction that occurs during roasting, grilling and baking of many food products
- (B) the connective tissue proteins (collagen, elastin and reticulum) and their conversion to gelatine
- (C) the action of the preservatives on the muscle fibres to help keep the product safe for longer
- (D) the reaction of the antioxidants used to prevent the fat in the food combining with oxygen and becoming rancid.

11. Which of the following is **not** a labelling requirement of food products having more than one ingredient?

[1 mark]

- (A) a list of ingredients
- (B) the name and address of the manufacturer, packager or importer
- (C) the nutritional content
- (D) a statement of the product's health benefits.

12. The Australian Guide to Healthy Eating encourages the consumption of:

[1 mark]

- (A) a low-fat, low-fibre diet
- (B) a variety of foods
- (C) simple carbohydrates instead of complex carbohydrates
- (D) a low-fat, high-protein diet.

13. HACCP is a management tool used in food processing environments to control which of the following?

[1 mark]

- (A) food quality issues
- (B) food safety issues
- (C) equipment cleaning and sanitation
- (D) nutritional issues.

14. The main purpose of blanching vegetables is to:

[1 mark]

- (A) heat the vegetable before canning
- (B) neutralise plant enzymes
- (C) increase moisture content
- (D) activate enzyme nutrient retention capability.

15. Which of the following is **not** a function of food additives?

[1 mark]

- (A) they enhance the flavour, colour and texture of processed foods
- (B) they enhance the nutritional quality of food
- (C) they increase the temperature of food
- (D) they prevent the growth of micro-organisms.

16. Which of the following instructions will **not** help to minimise food-borne illness?

[1 mark]

- (A) wash hands and cooking equipment frequently
- (B) select only the freshest foods when shopping
- (C) store all foods at room temperature
- (D) cook foods thoroughly.

17.

[1 mark]

For copyright reasons this question cannot be reproduced in the online version of this document, but may be viewed at
http://www.boardofstudies.nsw.edu.au/hsc_exams/hsc2006exams/pdf_doc/food_tech_06.pdf
(p.3 Q.1)

[Adapted from: Board of Studies New South Wales, 2006]

18. As fruits and vegetables age they give off carbon dioxide. To prolong the shelf life of these commodities, modern packaging is made from:

[1 mark]

- (A) cardboard to allow the product to breathe
- (B) barrier-specific packaging
- (C) hard plastic with holes
- (D) semi-permeable plastic.

19. Vinaigrette salad dressings are best described as:

[1 mark]

- (A) a temporary emulsion of oil, vinegar and mustard
- (B) a permanent emulsion of oil and vinegar
- (C) a dressing of egg yolk, mustard and oil
- (D) a mixture of lemon juice, sour cream and oil.

20. When thinly slicing herbs or green, leafy vegetables, the traditional cutting technique used is called:

[1 mark]

- (A) julienne
- (B) paysanne
- (C) brunoise
- (D) chiffonade.

END OF SECTION ONE

SEE NEXT PAGE

SECTION TWO—SHORT ANSWER

This section has **EIGHT (8)** questions. Attempt **ALL** questions.

Suggested working time 100 minutes [50 marks].

Question 1

Study the following graph, which compares the consumption of meat types in the years 1991/92 and 2007/08 and answer the questions which follow.

For copyright reasons this graph cannot be reproduced in the online version of this document, but may be viewed in its original form at http://esvc001418.wic023u.server-web.com/files/_system/Document/School%20DVD/RW%2008-013%20ACMF%20PRODUCTION%20SECONDARY.pdf.
(p.4. second graph)

[Graph adapted from: Australian Chicken Meat Federation (ACMF) Inc, n.d.]

(a) Identify two trends and suggest a reason for each trend:

[2 marks]

(b) Describe the nutritional contribution of meat to the Australian diet.

[2 marks]

(c) Explain how **two** sensory properties of meat are affected by heat in the cooking process.

[2 marks]

(d) Explain **four** key factors that should be considered when purchasing meat.

[4 marks]

(e) Define the term 'denaturation' and name two ways meat can be denatured.

[2 marks]

Question 2

Select a vegetable and name ways it could be cooked using wet and dry methods. How do the physical properties of the vegetable change with each cooking method and does this affect nutrient retention?

Name of vegetable: _____

Method of cooking	Changes in physical properties	Effect on nutrient retention
Wet method example:		
Dry method example:		

[6 marks]

Question 3

(a) Explain **two** important points to consider when separating egg whites.

[2 marks]

(b) Explain **four** factors which need to be considered when whisking egg whites to form peaks.

[4 marks]

SEE NEXT PAGE

Question 4

- (a) Identify and explain **two** factors that have led to an increase in the number of convenience foods available to the consumer.

[2 marks]

- (b) List and explain **three** effects that the increased availability of convenience foods has had on the health of individuals.

[3 marks]

Question 5

- (a) Select a fresh food commodity and list three methods that could be used to preserve it. Give an appropriate example for each method.

[3 marks]

Name of commodity: _____

- (b) For one of these methods explain how the physical properties of the food commodity are affected.

[2 marks]

Question 6

(a) What is meant by the term 'functional foods'?

[1 mark]

(b) Name one functional food and explain why consumers buy it.

[2 marks]

(c) Describe what is meant by a 'health claim'.

[1 mark]

Question 7

In recent times, food manufacturers have faced environmental, economic and social pressures from consumers. Explain how manufacturers have responded to these pressures.

[6 marks]

Environmental

Economic

Social

Question 8

You are required to produce a range of salads for a large-scale function.

- (a)** What aspects of food safety would need to be addressed during the preparation stage of the salads? Describe three key areas.

[3 marks]

- (b)** Describe three important presentation and quality considerations which need to be addressed when preparing salads for a function.

[3 marks]

END OF SECTION TWO

SECTION THREE—EXTENDED ANSWER

This section has **ONE (1)** compulsory question.

Suggested working time: 30 minutes [15 marks].

Question 11

Describe a food-related production or project where you used the technology process. To answer this question you should:

- Briefly outline your design brief and list the human and non-human resources used. [2 marks]
- Describe the target market. [1 mark]
- List the needs and values of the consumer and producer [2 marks]
- Identify and analyse how the key needs and values of the consumer and producer were addressed in the design and production processes. [5 marks]
- Use specific examples from the food-related production or project to explain in detail how you used effective management and communication skills to:
 - organise the production processes
 - manage issues that arose while working with others or in teams
 - ensure the quality requirements of the target market were met. [5 marks]

Blank lined paper with a large, faint, diagonal watermark reading "DRAFT" across the top left corner.

SEE NEXT PAGE

DRAFT

**END OF SECTION THREE
SEE NEXT PAGE**

SECTION FOUR—EXTENDED ANSWER (context-specific)

This section is divided into **THREE (3)** contexts. Indicate question answered.

Hospitality	Q. 12 ☐	OR	Q. 13 ☐
Product development	Q. 14 ☐	OR	Q. 15 ☐
Nutrition and health promotion	Q. 16 ☐	OR	Q. 17 ☐

Complete **ONE (1) CONTEXT ONLY**. There are **TWO (2)** questions per context. Answer **ONE (1)** question only.

Suggested working time: 30 minutes [15 marks].

THIS SPACE HAS BEEN LEFT BLANK INTENTIONALLY

HOSPITALITY**Question 12****Special function to demonstrate hospitality skills**

(a) Design an innovative **THREE (3)** course menu for one of the following listed functions:

- Youth Week to celebrate youth achievements
- promotion of local produce
- Melbourne Cup luncheon
- farewell meal for visiting overseas or regional students
- thank you to local businesses and community people to promote the school.

[4 marks]

(b) Write a cover sheet addressing the following:

- identify and describe the target market

[1 mark]

- explain the type of meal to be served (buffet or sit-down)

[2 marks]

- evaluate the planned menu, considering the functional, aesthetic, social and environmental factors that have influenced the decisions made.

[8 marks]

OR

Question 13**Meals in a moment**

For copyright reasons question 13 cannot be reproduced in the online version of this document.

[3 marks]

[4 marks]

[8 marks]

[Adapted from: Perraton et al., 2005]

SEE NEXT PAGE

PRODUCT DEVELOPMENT

Question 14

Healthy meal deals

As part of a healthy eating campaign you are required to design a healthy meal deal to be sold in a local primary school canteen for lunch. The meal is to cost no more than \$5.00 to produce.

In your response you will need to:

- Discuss **THREE (3)** important factors that need to be considered when designing a meal deal. [3 marks]
- List foods and drink that should be used in the healthy meal deal. Explain the choices you have made. [4 marks]
- Evaluate the functional, economic, social and environmental factors that have influenced your decisions. [8 marks]

OR

Question 15

Canteen Choices

Design a 'healthy—fresh is best' food wrap product to be sold at the local high school canteen using a variety of food products. The products are to be low cost.

In your response you will need to:

- identify and describe the target market [1 mark]
- describe the wrap and label [2 marks]
- describe the food packaging and promotion of this product [2 marks]
- explain your choices by considering the target market and the functional, economic, social and environmental factors that have influenced your decisions [8 marks]
- describe ways to evaluate the success of the product and identify product improvements. [2 marks]

NUTRITION AND HEALTH PROMOTION**Question 16****Promoting fruit and vegetables as healthy food choices**

As part of a healthy eating campaign you are to promote a food product that would appeal to people who under-consume fruit and vegetables.

In your response you will need to:

- Identify and describe a typical audience that under-consume fruit and vegetables. [1 mark]
- Describe **TWO (2)** promotional strategies for the product that would be effective with this audience. [4 marks]
- Evaluate strategies considering the nutritional, economic, social and environmental factors that have influenced your choice. [8 marks]
- Describe how you would evaluate the promotional strategies and any ongoing monitoring required. [2 marks]

OR

Question 17**Convenience and fast foods**

The media, busy lifestyles, increased incomes and the availability of convenience and fast food outlets have lead to increased levels of obesity in Australia amongst children and adults. Design a promotional strategy that communicates healthy choices to consumers purchasing products at an outlet.

In your response you will need to:

- Identify and describe the target audience that purchases convenience and fast food products. [1 mark]
- Describe a strategy to educate consumers about healthy food choices. [4 marks]
- Evaluate the planned strategy considering the nutritional, economic, social and environmental factors involved. [8 marks]
- Describe ways to evaluate the success of the strategy and identify improvements. [2 marks]

DRAFT

DRAFT

DRAFT

END OF PAPER

SEE NEXT PAGE

THIS PAGE HAS BEEN LEFT BLANK INTENTIONALLY

YOU MAY USE IT FOR PLANNING

ACKNOWLEDGEMENTS

SECTION ONE

- Question 17** Adapted from: Board of Studies New South Wales. (2006). *Food Technology: 2006 Higher School Certificate Examination*. Retrieved April, 2008, from http://www.boardofstudies.nsw.edu.au/hsc_exams/hsc2006exams/pdf_doc/food_tech_06.pdf.

SECTION TWO

- Question 1** Graph adapted from: Australian Chicken Meat Federation (ACMF) Inc. (n.d.). [Graph: Meat consumption—Australia]. Retrieved April, 2008, from <http://esvc001418.wic023u.server-web.com/files/system/Document/School%20DVD/RW%2008-013%20ACMF%20PRODUCTION%20SECONDARY.pdf>.

SECTION FOUR

- Question 13** Adapted from: Perraton, G., et al. (2005). *Food and technology 1*. Milton, Qld: John Wiley & Sons, p. 191.